

Waterborne Polyurethane Hardener

NO.: MSDS202275911548

Revision Date: 14.01.2022

1) Chemical product and Company Identification

Product Chinese name: 水性聚氨酯固化剂

Product English name: Waterborne Polyurethane Hardener

End Uses: Polyurethane curing agent for coatings or adhesives in industrial or commercial applications

Manufacturer/Supplier:

Foshan City Nanhai Wanlei Building Paint CO., LTD.

No.12 Songxia Industrial Area, Nanhai, Foshan, China

Telephone: +86-757-81760676

Fax: +86-757-85227712

E-mail: rita@fswanlei.com

Emergency telephone number: +86-13823401617

2) Hazards Identification

Emergency overview

Colorless Transparent Liquid

GHS Classification

Inhaled 4-H332, Harmful if inhaled

Skin Sens.1-H317, May cause an allergic skin reaction.

Eye Irrit. 2-H319, Cause eye irritation.

3-H412, Hazardous to the aquatic environment

Label elements

Hazard pictograms



Signal word

Caution

Hazard statements

H332: Harmful if inhaled

H317: May cause an allergic skin reaction.

H319: Cause eye irritation.

H412: Hazardous to the aquatic environment

Precautionary statements

P261 Avoid breathing dust/fume/gas/mist/vapours/spray

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P273 No discharge into the environment

P302+P352 IF ON SKIN: Wash with plenty of water

P312 Call a POISON CENTER/doctor if you feel unwell.

P378 Suitable fire extinguisher: carbon dioxide (CO2), foam, fire extinguishing powder

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P403+P233 Store in a well-ventilated place. Keep container tightly closed.
P501 Dispose of contents/container to an approved waste disposal plant
in accordance with local/regional/national/international regulation.

3) Composition/Information on IngredientSubstances ☐ Mixture ☒

Component	CAS No.	Concentration%
Water dispersible isocyanate	75-13-8	100
1,6-hexamethylene diisocyanate	822-06-0	0.4 MAX

4) First-aid Measures**Description of first aid measures****General advice**

If symptoms persist, call a physician.

Inhalation

If breathed in, move person into fresh air. If not breathing, give artificial respiration. If symptoms persist, call a physician.

Skin contact

Wash off with soap and plenty of water. If symptoms persist, call a physician.

Eye contact

Flush eyes with water as a precaution. If symptoms persist, call a physician.

Ingestion

Rinse mouth with water. Call a physician.

Most important symptoms and health effects

No further relevant information available.

Advice for the protection of rescuers

No further relevant information available.

Special advices for doctors

Treat symptomatically.

5) Fire Safety Measures**Suitable extinguishing media**

Use water, foam, dry chemical or carbon dioxide.

Unsuitable extinguishing media

No further relevant information available.

Special hazards

No further relevant information available.

Precautions and measures for fire extinguishing

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Wear self contained breathing apparatus for fire fighting if necessary.

6) Accidental Release Measures**Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment as required.

Environmental precautions

Should not be released into the environment. For more ecological information, please refer to section 12.

Methods and material for containment and cleaning up

Contain spillage, and then place in container for disposal according to local regulations.

Preventive measures to prevent the occurrence of secondary disasters

See Section 7 for information on safe handling.

7) Handling and Storage**Handling**

Avoid contact with eyes and skin. Avoid ingestion and inhalation. Wash hands before breaks and immediately after handling the product.

Storage

Store away from incompatible materials described in Section 10. Store in a dry, cool and well-ventilated place. Keep the container closed.

8) Exposure Control and Individual Protection**Occupational exposure limits**

Chemical name	Reference	Type	Value
1,6-hexamethylene diisocyanate	CN OEL	STEL	0.03mg/m ³

备注: Reference to 《GBZ 2.1-2007》

Engineering controls

Use with local exhaust ventilation. Ensure adequate ventilation, especially in confined areas.

Personal protective equipment**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

Eye/Face protection

Use equipment for eye protection tested and approved under appropriate government standards

Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Hand protection

Protective gloves.

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9) Physical and Chemical Properties**Information on basic physical and chemical properties**

Appearance and character	Colorless Transparent Liquid
Odor	Weak odor
Ph-value	No data available
Melting point/ Freezing point	No data available
Boiling point/Initial boiling point/Boiling range	<300℃
Flash point(closed cup)	≈160℃
Upper/lower flammability or explosive limits	
Lower	No data available
Upper	No data available
Vapour pressure	No data available
Vapour density	No data available
Density/ Relative density	No data available
Solubility	Insoluble in water/15℃
Partition coefficient(n-octanol/water)	No data available
Auto-ignition temperature	No data available
Decomposition temperature	No data available
Flammability (solid, gaseous)	Not applicable
Evaporation rate	No data available
Odor Threshold	No data available
Explosive properties	No data available
Oxidizing properties	The substance does not belong to oxidizing substances
Viscosity	
Dynamic	≈ 800cp/25℃
Kinematic	No data available
Other information	No information available.

10) Stability and Reactivity

Stability	Stable at normal temperatures and storage conditions
Incompatible materials	Strong oxidizing agents.
Hazardous reactions	None under normal processing.
Conditions to avoid	It is dangerous to react with alcohols and amines in closed containers due to the rise of pressure and explosion

11) Toxicological Information

Acute toxicity Based on available data, the classification criteria are not met.

Water dispersible isocyanate

Oral : LD50,Rat: >2000mg/kg

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Dermal: Skin contact may cause irritation and allergy.

Inhalation: LC50, Rat, female: >390mg/m³, 4h

Skin corrosion/irritation Slight irritation, long-term contact with the skin, may have brown spots or irritating reactions.

Serious eye damage/eye irritation If this product is splashed into the eyes, it will stimulate and temporarily damage the eyes.

Respiratory or skin sensitization Irritating to respiratory tract and sensitizing to guinea pigs.

Germ cell mutagenicity Based on available data, the classification criteria are not met.

Carcinogenicity Based on available data, the classification criteria are not met.

Reproductive toxicity Based on available data, the classification criteria are not met.

12) Ecological Information

Eco-toxicity Based on available data, the classification criteria are not met.

Water dispersible isocyanate

Acute fish toxicity: LC50, Zebrafish: >28.3 mg/L, 96h.

Acute algal toxicity: EC50, *Scenedesmus spinosa*: >100mg/L, 72h.

Acute bacterial toxicity: EC50, activated sludge: >10000mg/L.

Persistence and degradability : 2%, 28d, Solids are not easy to degrade.

Bioaccumulative potential No data available.

Mobility in soil No data available.

13) Disposal Considerations**Waste treatment methods****Recommendation - Waste from Residues I Unused Products**

Follow all local, state, and federal regulations for disposal of waste chemicals

Recommendation - Contaminated Packaging

Follow all local, state, and federal regulations for disposal of waste chemicals.

14) Transport Information

Land (Not dangerous goods)

IATA (Not dangerous goods)

IMDG (Not dangerous goods)

UN-Number Not Regulated (Not dangerous goods)

Notes: non dangerous goods. Keep dry; The temperature shall not be higher than + 50 °C and lower than - 15 °C, Stay away from food, acids and alkalis

15) Regulatory Information

Rules for classification and labeling of chemicals (GB 30000.2-2013 - GB30000.29 -2013)

Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

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General rules for preparation of chemical safety data sheet GB/T16483-2008
Guidance on the compilation of safety data sheet for chemical products GB/T 17519-2013
Law of the Peoples Republic of China on the Prevention and Control of Occupational Diseases
Regulation on the Safety Administration of Dangerous Chemicals
The Catalogue of Hazardous Chemicals
Inventory of the Existing Chemical Substances in China
China Catalog of Strictly Restricted Toxic Chemicals
Inventory of the Existing Chemical Substances in China
Catalogue of Toxic Chemicals Strictly Restricted from Import and Export in China
National Occupational Health Standards of the Peoples Republic of China
General rule for classification and hazard communication of chemicals
Classification and code of dangerous goods
Measures for Environmental Management of New Chemical Substances
List of the First Batch of Hazardous Chemicals under Key Supervision
List of dangerous chemicals under the second batch of key supervision
List of dangerous goods
Packing symbol of dangerous goods
Regulations on Safe Use of Chemicals in Workplaces
The proposal workplaces safe use of chemicals(No.177)
Recommendations on the TRANSPORT OF DANGEROVS GOODS

16) Additional Information

The contents and formals of the SDS are prepared in accordance with Regulations GB 16483-2008 and GB/T17519- 2013.

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